(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

IV Semester, April, 2024

Paper Code: CIC-206 **Subject: Theory of Computation**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1			
a.	Define Regular expression.	2.5	CO1
b.	Prove that Regular languages are closed under intersection.	2.5	CO1
c.	Define Ambiguity in CFL with the help of an example.	2.5	CO2
d.	State and prove pumping Lemma for Context Free Grammar.	2.5	CO2
2			
a.	Design NDFA for a language over input alphabet {0,1} accepting strings in which number of 1s are divisible by either three or five.	5	CO1
b.	Design a Push Down Automata for L(G)={$a^n b^n$, n>=0}	5	CO2
3			
a.	Convert the following NFA to DFA State 0 1 q0 q0, q1 q0 q1 q1 q2 q2 q1, q2 q0 q0 is the initial state and q2 is the only final state.	5	CO1
b	Define LL(K) parsing in detail with the help of an example.	5	CO2
4			
a.	Find the regular expression equivalent to following FA State a b q1 q1 q2 q2 q3 q2 q3 q2 q1 q1 is the initial state and q1 is the only final state	5	CO1
b.	Define Normal Forms for Grammar. Convert the following CFG in to Chomsky normal form S→aAb| bBa| Ɛ A→abc|aAB B→cba	5	CO2